

Concept Question 8.1 Star formation and fragmentation

Consider two molecular clouds with the same total mass and temperature. Cloud A has nearly uniform density, while cloud B contains several dense clumps produced by turbulence.

- a) Which cloud would begin forming stars first? Explain using the dependence of free-fall time on density.
- b) During the nearly isothermal collapse phase, which cloud would you expect to produce more low-mass fragments? Why?

Problem 8.2 Turn-over point in thermal radio spectrum

Calculate the turn-over frequency between the optically thick and the optically thin part of the radio emission of a nebula with $N_e \approx N_p = 10^2 \text{ cm}^{-3}$, $T = 10\,000 \text{ K}$ ($g_{\text{ff}} = 1$), and a volume of 1 pc^3 (cube with length of 1 pc).

Hint: The optically thick part of the spectrum is given by the Rayleigh-Jeans part of the Planck curve. For the optically thin part you can assume thermal bremsstrahlung with emission coefficient:

$$j_\nu = \frac{5.44 \times 10^{-39}}{T^{1/2}} g_{\text{ff}} N_i N_e e^{h\nu/kT} \quad (1)$$

where N_i and N_e are the number densities of ions and electrons, respectively.

Problem 8.3 Two-Level Excitation in the ISM (C ii, O i, CO)

We treat each transition as an effective two-level system with lower level 1 and upper level 2. Assume **no external radiation field** (ignore stimulated emission and radiative absorption). Collisions with a dominant collider of number density n and kinetic temperature T drive excitation and de-excitation.

a) General two-level results

1. Write the statistical equilibrium equation for the upper level in steady state and show that

$$\frac{N_2}{N_1} = \frac{nQ_{12}}{A_{21} + nQ_{21}}. \quad (2)$$

2. Using detailed balance,

$$\frac{Q_{12}}{Q_{21}} = \frac{g_2}{g_1} \exp\left(-\frac{\Delta E}{k_B T}\right), \quad (3)$$

show that in the high-density limit ($n \gg N_{\text{crit}}$) the population ratio tends to the LTE (Boltzmann) value.

3. Define the *critical density* and show that

$$N_{\text{crit}} = \frac{A_{21}}{Q_{21}}. \quad (4)$$

b) Numerical applications

For each case below (1) compute the LTE ratio $(N_2/N_1)_{\text{LTE}}$; (2) for $n = 10^2, 10^4, 10^6 \text{ cm}^{-3}$, when can the transition be classified as LTE and when as sub-thermal?

Case 1: C ii 158 μm

$$A_{21} = 2.3 \times 10^{-6} \text{ s}^{-1}, \quad \frac{\Delta E}{k_B} = 91.2 \text{ K}, \quad \frac{g_2}{g_1} = 2, \quad T = 100 \text{ K}.$$

Adopt $N_{\text{crit}} \approx 6.1 \times 10^3 \text{ cm}^{-3}$ and use $Q_{21} = A_{21}/N_{\text{crit}}$.

Case 2: O i 63 μm

$$A_{21} = 8.7 \times 10^{-5} \text{ s}^{-1}, \quad \frac{\Delta E}{k_B} = 227.7 \text{ K}, \quad \frac{g_2}{g_1} = \frac{3}{5}, \quad T = 200 \text{ K}.$$

Adopt $N_{\text{crit}} \approx 5 \times 10^5 \text{ cm}^{-3}$ and use $Q_{21} = A_{21}/N_{\text{crit}}$.

Case 3: CO $J=1 \rightarrow 0$ (115.27 GHz)

$$A_{10} = 7.2 \times 10^{-8} \text{ s}^{-1}, \quad \frac{\Delta E}{k_B} = 5.53 \text{ K}, \quad \frac{g_1}{g_0} = 3, \quad T = 10 \text{ K}.$$

Adopt $N_{\text{crit}} \approx 2.0 \times 10^3 \text{ cm}^{-3}$ and use $Q_{10} = A_{10}/N_{\text{crit}}$.

c) Interpretation

1. Which of the three lines is easiest to thermalise in typical molecular cloud gas, and why?
2. Explain why C II 158 μm can be bright in diffuse gas, whereas O I 63 μm usually requires dense, warm gas.
3. For CO(1–0), explain qualitatively why real clouds can show bright emission even when $n < N_{\text{crit}}$ (hint: optical depth / radiative trapping lowers the *effective* critical density).